

There's a Time and a Face: The Time Course of Composite Face Processing

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Many decades of face processing research have converged on the idea of faces being special objects (Farah, Wilson, Drain, & Tanaka, 1998). One demonstration that highlights this is the inversion effect, where presenting a face in an inverted configuration reduces recognition and discrimination performance, more so than for other objects (Thompson, 1980; Yin, 1969). Another demonstration comes from the composite face effect, where the illusion of a unique face emerges upon aligning the top half of one face with the bottom half of another (Young, Hellawell, & Hay, 1987). In this arrangement, recognition for each half is poorer compared to when the composite is misaligned (Rossion & Boremanse, 2008; Young et al., 1987). The to-be-ignored half inevitably interferes with the attended half, described as a failure of selective attention (Richler, Palmeri, & Gauthier, 2012). This provides one of the many operational definitions of holistic processing (Richler et al., 2012).

One issue with studying holistic processing is that there are many different empirical measures and mechanisms that have been investigated. For instance, in the part-whole task (Tanaka & Farah, 1993) enhanced recognition performance for a face-part in the context of a whole face, as opposed to in isolation, is also taken as evidence for holistic processing. Even though this measure may tap into an alternative hypothesis – that the whole is greater than the sum of its parts (Richler et al., 2012). This is only one example of over a dozen tasks that claim

to measure the same construct. Some of these tasks even disagree with each other; as such, there is no unified theory of holistic processing (see Richler et al., 2012 for a review).

When considering the composite face paradigm alone, despite the large body of evidence to support holistic processing (Richler & Gauthier, 2014), we still find some conflicting results. Fitousi (2015) assessed two notions of holistic processing by testing for integrality with classic Garner tasks (Garner, 1974) and supercapacity (consistent with coactive processing) with a redundant target task (Townsend & Nozawa, 1995). He found that composite faces exhibited neither integrality nor supercapacity, though whether these tasks tap into the same construct remains disputed (Richler, Palmeri, & Gauthier, 2015, Richler et al., 2012; Little, Nosofsky, Donkin, & Denton, 2013). Additionally, Cheng, McCarthy, Wang, Palmeri, and Little (2018) examined the architecture of composite face processing with Systems Factorial Technology (SFT; Townsend & Nozawa, 1995) and found that regardless of orientation or alignment, performance was largely indicative of a mixture of serial and parallel processing rather than coactive. These inconclusive findings make it difficult to predict how composite faces will perform in new contexts. Here we hope to better understand the composite face effect by taking a closer look at the time course of composite face processing.

There are few studies that have assessed the complete time course of composite face processing. Hole (1994) did find holistic processing at a short presentation time (80 ms) but not at a longer presentation time (2000 ms). Though, one should take this finding lightly as holistic processing was inferred from an inversion effect within a partial design, meaning any difference in discriminability could be confounded by a response bias (see Gauthier & Bukach, 2007). Cheng et al. (2018) used a more rigorous complete design to look at mean RT patterns and how

they map to processing architectures. Nevertheless, a more direct approach involves systematically limiting the amount of time to process a composite face and mapping the resulting changes in accuracy over time. The most compelling example of this comes from Richler, Mack, Gauthier, and Palmeri (2009) who assessed the time course of encoding by limiting the presentation time of a study or a test face. They found that at the shortest presentation time (50 ms) holistic processing was present for both upright and inverted faces, as indicated by congruency effects, through to the longest presentation time (1800 ms). We intend to apply their methodology in the current design by limiting the time available for processing as well as the time available to respond. Holistic processing will likewise be inferred from congruency effects, but a congruency x alignment interaction may also be assessed. Taken together, this will allow us to directly assess whether there are interference effects present from the to-be-ignored face half, as well as when in the time course these effects may occur.

Method

Participants

Three participants from the University of Melbourne will complete 15 one-hour sessions of the experiment, providing a large number of observations for each item condition. This will allow assessment of the time course of composite face processing across multiple interference conditions at the individual participant level; consequently, each participant is a replication of the entire experiment (Smith & Little, 2018) and each response for each item at each deadline is measured with high precision. Participants may be recruited from the authors' laboratory and

from the University of Melbourne community. The latter participants will receive \$15 per session for their testing. All participants have normal or corrected-to-normal vision. Human testing has been approved by the University of Melbourne Human Ethics Research Committee (Approval number: 1034866.16).

Stimuli and Apparatus

The stimuli will be composite faces consisting of a top and bottom face half. To create the top and bottom face halves, we selected a set of 54 *base faces* from Kayser (1997; see also Steyvers, 1999). We then created 20 *morph sets* by selecting four base faces (A, B, C, and D) at random and morphing them together using a field morphing technique (Steyvers, 1999) such that each morph set was a 3 x 3 face set with variation between A and B forming one dimension and between C and D forming the other dimension (see Figure 1). For instance, the centre face has equal proportions of A, B, C, and D. On each trial, a top half and a bottom half will be randomly drawn from the *corner* items of two morph sets to create a study face. Hence, from the 20 morph sets, there are 80 potential top halves and 80 potential bottom halves making 640 unique faces.

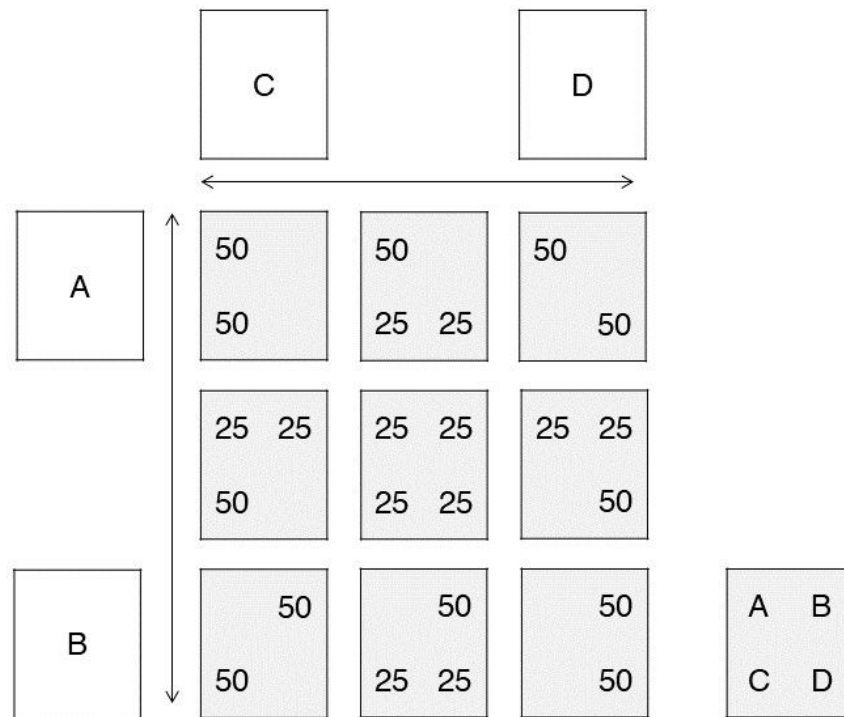


Figure 1. Morph space. 3 x 3 face set created by field morphing four base faces, A, B, C, and D, together along an A and B dimension, and a C and D dimension. The numbers denote the proportion of each base face contributing to creating each new face.

On each trial, a study face will be randomly generated and followed by a test face. Test faces will also be composite faces with a top half drawn from the same morph set as the top of the study face and the bottom half drawn from the same morph set as the bottom. The top and bottom can be the same (S) or different from the study face. If different, then the magnitude of the difference can be of low discriminability (L) or high discriminability (H). In terms of the morph set, a low change would be one step on each dimension of the morph space, while a high change would be two steps on each dimension of the morph space (Figure 1). All combinations

of S, L, and H will be used; hence, the test composite face can be one of nine options (Figure 2).

		Top Face Half		
		S	L	H
Bottom Face Half	S	S	L	H
		S	S	S
	L	S	L	H
		L	L	L
	H	S	L	H
		H	H	H

Figure 2. Test composite face combinations. The top and bottom can be the same (S) or different from the study composite face. Differences can either be of low discriminability (L) or high discriminability (H). All nine combinations will be used as different test face conditions.

Participants will be tested in a dark room with the stimuli presented on a 1280 x 1024 resolution Dell 22" monitor. Each composite face will be 400 x 340 pixels, which at a viewing distance of 60 cm, will subtend a visual angle of $6.68^\circ \times 5.67^\circ$. A gap of 10 pixels will subtend a visual angle of 0.17° and will provide a smoother transition between the morphed halves. A Gaussian blur will be applied to the edges of each face. RTs will be recorded on a calibrated RT box (Li, Liang, Kleiner, & Lu, 2010).

Follow-up experimental designs will be the same but stimuli will be presented in upright misaligned, inverted aligned, and inverted misaligned configurations.

Design and Procedure

Participants will complete one training and 14 experimental sessions over several weeks. The training session will have six blocks of 78 trials with blank face stimuli, where the goal is to practice responding within 300 ms of a response signal, followed by four blocks of 78 trials making easy face discriminations (using only S/S and H/H trials). Each experimental session will consist of ten blocks of 78 trials, equating to 780 trials. Trials will be divided into 130 trials for each of the three composites where the top half remains the same (S/S, S/L, S/H), and 65 trials for each of the six composites where there is a change in the top face (L/S, L/L, L/H, H/S, H/L, H/H). Dividing the trials up in this way will equate the number of same and different trials. The order of trials will be randomized within each session. Participants will be told to prioritise responding on time (i.e., within 300 ms of the response signal) rather than responding accurately.

A fixation cross will be presented for 500 ms at the start of each trial before a study face is presented for 500 ms (Figure 3). After a blank screen for 500 ms, a test face will appear for 500 ms. Once the response signal is presented, participants will have to indicate whether the top half of the face was the same or different with the left and right buttons on the RT box respectively. The response signal, a set of XXXs appearing above and below the face, will appear at one of six different lags: 50, 100, 200, 400, 800, and 1800 ms. Feedback will be provided for the accuracy of their response (i.e., “CORRECT” or “WRONG”) for 250 ms followed by a blank screen for 400 ms. Feedback will then be provided for the timing of their

response for 500 ms followed by a blank screen for 300 ms. Participants will be told if it was within 100 ms after the response signal (i.e., “TOO FAST”), if it was 300 ms or longer after the response signal (i.e., “TOO SLOW”), if they have responded before the response signal (“PLEASE WAIT FOR THE RESPONSE SIGNAL”), or if it was an ideal response (i.e., “ON TIME”). Participants will be allowed to take short breaks in between blocks.

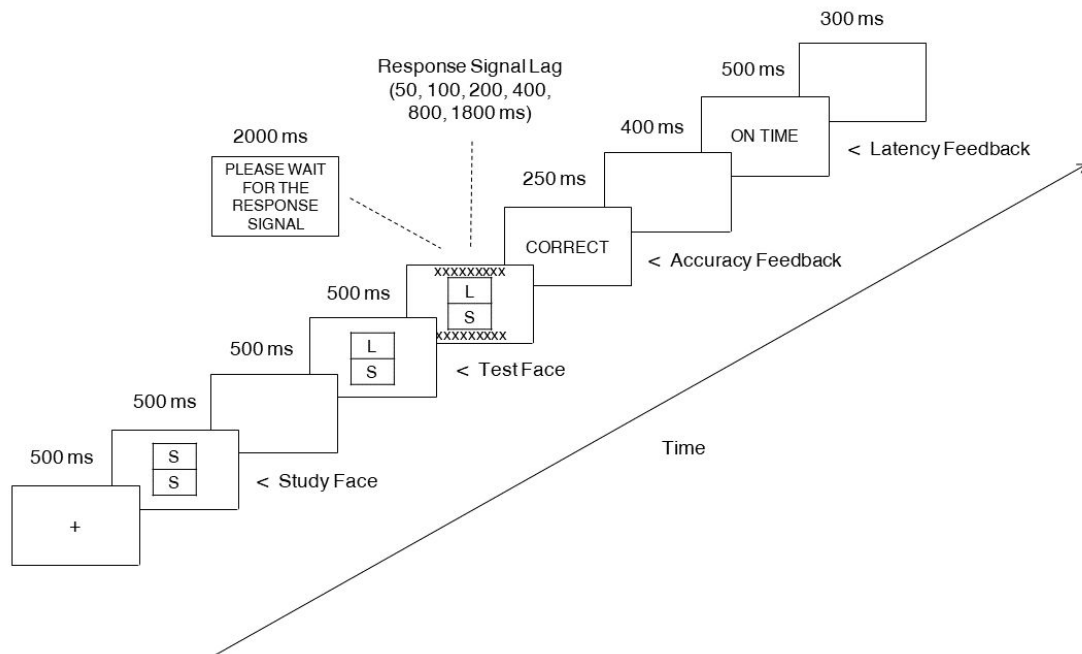


Figure 3. Single trial example. An accurate, on time response is illustrated when a low discriminability change has occurred in the top face half. XXX = the response signal.

Data Analysis

A cued-response speed-accuracy trade-off (SAT) method will be used to investigate the time course of possible composite face effects. The requirement that participants must immediately respond to a signal will interrupt the retrieval of the study face from memory at

various stages. Varying the response signals between 50 and 1800 ms will allow us to measure accuracy and sensitivity at these times and generate SAT functions of the growth of accuracy (Ratcliff, 2006).

To compute sensitivity and bias, we will use the differencing signal detection model introduced by Sorkin (1962). Since we are concerned solely with detection of any change (rather than the arbitrary direction of that change), the change strength is restricted to be positive and there is a single criteria separating the noise and change signal distributions. Consequently, two parameters will be used: the sensitivity (d') and the criterion (c). Sensitivity will be plotted for each condition across response signals. We will fit exponential growth functions to capture the properties of the SAT functions for comparison across conditions:

$$d'(t) = d'_a(1 - \exp(-(t - T_0)/\tau)), \quad \text{for } T_0 > 0 \quad (1)$$

Three parameters of interest will be assessed from these functions: the asymptotic accuracy which describes the maximum sensitivity (d'_a), the intercept which describes the time at which information begins accumulating above chance (T_0), and the rate of this accumulation (τ).

Models will be fit to our data using the above three parameters using both Maximum Likelihood and Bayesian estimation procedures implemented in JAGS, STAN or Matlab (both procedures will be used for pedagogical reasons).

Sensitivity will be computed across conditions in the following way:

- 1) First, we expect to find that sensitivity is increased across the entire time course as the strength of the change in the top face changes. To assess this, we will compare d' computed from S/S and L/S to d' computed from S/S and H/S.
- 2) Second, we expect a change in the bottom half with no corresponding change in the top half to decrease asymptotic sensitivity, reflected by an increase in false alarms. To assess this, we will compare d' computed from S/S vs S/L to d' computed from S/S vs S/H. We make no specific predictions regarding the growth or start parameters of the SAT function.
- 3) Third, we expect a change in the bottom half of the face that co-occurs with a change in the top half to increase asymptotic sensitivity reflected by an increase in the hit rate. To assess this, we will compare d' computed from the following: S/S vs L/L, S/S vs L/H, S/S vs H/L, and S/S vs H/H, with d' expected to increase across those conditions if there is interference. If there is no interference from the bottom face, then d' should be equivalent between the L/L and L/H conditions and between the H/L and H/H conditions.

To draw inferences about differences between conditions, we will fit models with parameters that are free to vary between conditions and compare them using G-squared tests to models with parameters constrained to be equal across conditions. For Bayesian tests, we will use posterior comparison along with model comparison using DIC or WAIC.

Additional model variants will be explored for mechanisms such as anticipation for the response signal. At short time lags (e.g. 100 ms) we expect participants to make decisions based on partial information, whilst at longer lags (1800 ms) we expect mainly accurate terminated

decisions, where participants are simply waiting for the response signal (Ratcliff, 2006).

Variability in RTs within the 100-300 ms window after each time lag will also be considered.

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